

AUTOMATED SEGMENTATION OF PHANTOM CALIBRATION RODS IN HETEROGENEOUS CT SCANS USING DEEP LEARNING

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ABSTRACT

Accurate segmentation of phantom calibration rods is important for quantitative computed tomography (CT) workflows, yet existing approaches are often manual, time consuming or sensitive to acquisition variability. This work applied a deep learning method for automated binary segmentation of phantom calibration rods in heterogeneous CT scans. The task was defined as foreground-versus-background segmentation of all rods as a single class, using the nnU-Net v2 framework. The training dataset included 60 labelled CT scans spanning knee, chest, abdomen, and phantom-only asynchronous acquisitions, with further variability in tube energy, scanner vendor, reconstruction kernel, table height, acquisition site, and artifact status. Performance was evaluated using the Dice Similarity Coefficient (DSC) and intersection-over-union (IoU). In testing, the model achieved a mean DSC of 0.872 and a mean IoU of 0.781, indicating that this nnU-Net model can provide accurate and scalable rod segmentation despite substantial acquisition heterogeneity.

Index Terms— Computed Tomography (CT), Segmentation, Phantom Calibration, Deep Learning (DL), nnU-Net

1. INTRODUCTION

Deep learning (DL) is widely used in image analysis because multilayer neural networks learn hierarchical representations directly from data, reducing reliance on hand-crafted features and enabling strong performance in complex visual tasks [1]. In medical imaging, segmentation is a central DL application, enabling quantitative analysis, diagnosis, treatment planning, and downstream clinical assessment [2, 3, 4].

Among segmentation architectures, U-Net and its variants have become especially influential due to their encoder-decoder structure with skip connections, which supports both contextual feature learning and accurate localization [5, 4]. Review studies identify U-Net-based models as a dominant approach for medical image segmentation across modalities such as CT, MRI, ultrasound, PET, and X-ray. Building on this paradigm, nnU-Net has emerged as a strong

automated baseline due to its self-configuring framework, which adapts preprocessing, network architecture, training, and post-processing to dataset-specific properties, enabling robust performance across diverse biomedical segmentation tasks [4, 2].

Calibration phantoms are important in CT imaging, particularly in the calculation of volumetric bone mineral density (vBMD) [6]. Accurate BMD measurement requires calibration phantoms that mimic the attenuation properties of bone and have known density equivalents. In addition, CT acquisition varies across scanners, imaging parameters, and reconstruction algorithms, introducing variability in Hounsfield Unit measurements. Calibration phantoms and appropriate calibration procedures reduce this inter-scanner variability and ensure consistent and accurate quantitative measurements [7]. As a result, accurate identification and segmentation of phantom structures are essential components of quantitative CT workflows.

Given the inherent variability in CT acquisition, a segmentation framework that can adapt to heterogeneous imaging conditions is required. The nnU-Net is well suited to this setting as it automatically configures preprocessing, architecture, and training based on dataset characteristics [2]. Benchmarking studies further show that U-Net-based methods within the nnU-Net framework remain among the most reliable approaches when evaluated against strong baselines [8]. Therefore, the present work prioritizes robustness and reproducibility over architectural novelty, motivating the use of nnU-Net as a standardized baseline.

The objective of this work is to develop and evaluate a DL-based method for automatic binary segmentation of phantom calibration rods in heterogeneous CT scans. The central hypothesis is that a self-configuring nnU-Net pipeline can achieve accurate segmentation despite variability in scanner, acquisition protocol, and imaging artifacts, thereby providing a scalable and reproducible alternative to manual segmentation.

2. METHODS

2.1. Dataset

The labelled dataset comprised 60 CT scans collected from four acquisition groups: 20 CT knee images (mean age = 32, female = 13, male = 7), 10 CT chest images (mean age = 28, female = 4, male = 6), 10 CT abdomen images (mean age = 33, female = 6, male = 4), and 20 asynchronous phantom scans without anatomy. These scans were assembled from multiple internal datasets collected at the University of Calgary’s Centre for Mobility and Joint Health and were selected to capture variability in acquisition conditions, and phantom type. The knee scans were taken using an 80/140 fast switching kVp scanner (Siemens GE Revolution HD GSI). Chest and abdomen scans were taken on the same GE Revolution HD GSI but were acquired using a single energy of 120 kVp . The asynchronous scans were taken at a range of 80–140 kVp and at table heights ranging from 0–505 mm (Canon, GE, and Siemens scanners). All images were stored in NIfTI format, and manual ground-truth segmentations were produced using ITK-SNAP (V 4.4.0). Segmentations were created by three separate annotators, using the phantom investigational user guide [9].

Dataset variability was preserved because robust segmentation in practice requires handling acquisition shifts rather than assuming narrowly standardized input data. The final dataset therefore reflects a realistic range of CT appearances rather than a single-protocol cohort, as illustrated in Figure 1.

2.2. Data Organization and Annotation

The project followed the nnU-Net raw data structure as defined in the official GitHub repository [10, 2], with CT images stored in `imagesTr` and binary segmentation masks stored in `labelsTr`. Since the final task was binary segmentation, the dataset metadata were configured with one imaging channel (CT) and two labels: `background = 0` and `rod = 1`.

2.3. Model Architecture

The nnU-Net v2 framework was selected for this segmentation model (see Figure 2). This framework is not a single fixed architecture; rather, it is a self-configuring pipeline that derives preprocessing, network topology, patch size, batch size, resampling strategy, training schedule, and post-processing from dataset properties [2]. This makes it especially appropriate for the current project, where scans differ in anatomy, scanner vendor, reconstruction settings, and artifact burden. The framework has also become a strong reference standard in medical image segmentation due to its out-of-the-box performance and broad applicability [4, 8].

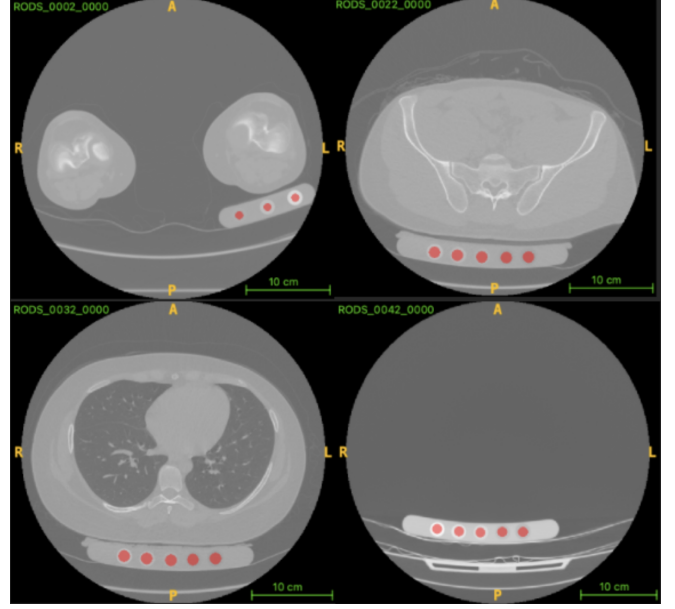


Fig. 1. Representative examples from the heterogeneous CT dataset used for phantom rod segmentation. The panels illustrate the range of imaging contexts included in the study, such as knee (top left), abdomen (top right), chest (bottom left), and asynchronous phantom-only (bottom right) scans. Corresponding manual binary annotations are overlaid to show the target foreground class.

2.3.1. Preprocessing and Augmentation

Preprocessing was performed using the default planning and preprocessing pipeline of nnU-Net v2. Dataset fingerprint extraction analyzed image size, voxel spacing, intensity distribution, and class balance to determine dataset-specific preprocessing and training parameters, including resampling target, patch size, batch size, and network architecture [2].

The experimental planning stage generated configurations for 2D, 3D low-resolution, 3D full-resolution, and 3D cascade training, all of which are present in the resulting plans file. The dataset exhibited near-isotropic voxel spacing (median: $0.703 \times 0.625 \times 0.703$ mm), and the median image size after resampling was $512 \times 488 \times 512$ *voxels*. Given the near isotropic nature of the data, the 3D full-resolution configuration was selected for training, as it is the recommended configuration for volumetric CT data without significant anisotropy. The 2D configuration is generally used for anisotropic volumetric data, and the 3D cascade (low-resolution followed by full-resolution) is only triggered when the patch size of the 3D full-resolution U-Net covers less than 12.5% of the median image shape [2], which is not the case with our data. In our data, the 3D full-resolution patch size of 128 voxels covered 25.0%, 26.2%, and 25.0% of the median image extent in the y , x , and z dimensions respectively, meeting the minimum per-dimension coverage threshold used by

nnU-Net to determine whether the cascade pipeline is necessary. As all three dimensions met this threshold, the cascade configuration was not triggered.

To improve model robustness and generalization, data augmentation was applied during training. Augmentation strategies in the nnU-Net v2 framework included rotation, scaling, Gaussian noise, Gaussian blur, brightness and contrast adjustments, gamma correction, simulation of low-resolution images, and mirroring. These augmentation techniques are commonly used in medical image segmentation to address limited dataset size and improve model invariance to variations in image acquisition and quality.

2.3.2. Training Strategy

Training was performed with nnU-Net v2 using PyTorch on a GPU-based environment (NVIDIA A100 80gb) with *five-fold cross-validation* used as the default validation strategy. In each of the five folds, nnU-Net automatically splits the data splits using 80% for training and 20% for validation. In our data, 48 of the 60 labelled scans of our training data are used for training, and 12 are used for validation. The use of five folds is consistent with recommendations for more reliable segmentation benchmarking in medical imaging [8].

The model was trained using the default nnU-Net training schedule, which runs for 1000 epochs using stochastic gradient descent (SGD) with Nesterov momentum ($\mu = 0.99$) and an initial learning rate of 0.01, decayed over training according to a polynomial learning rate policy [2]. The loss function used in nnU-Net sums the Dice loss and cross-entropy loss, which is appropriate for binary medical segmentation because it balances region overlap quality with voxel-wise classification performance. This combination is also widely used in recent medical image segmentation studies [11].

2.3.3. Evaluation Metric

Model performance was assessed using the Dice Similarity Coefficient (DSC) and the intersection-over-union (IoU), which quantify overlap between prediction and ground truth. DSC and IoU are widely used in medical image segmentation as it provides an interpretable measure of agreement between automated predictions and ground-truth annotations [3, 4, 12]. Inference was performed on a set of 21 unseen test images (7 knee, 7 chest, 7 abdominal) and test ground truth labels to calculate the DSC and IoU for the test set.

3. RESULTS

The nnU-Net-based rod segmentation pipeline achieved strong overlap-based performance under the current experimental setup. Across cross-validation folds the mean DSC was 0.904 and the mean IoU was 0.827, with a best observed DSC of 0.943 and a best IoU of 0.892, indicating accurate

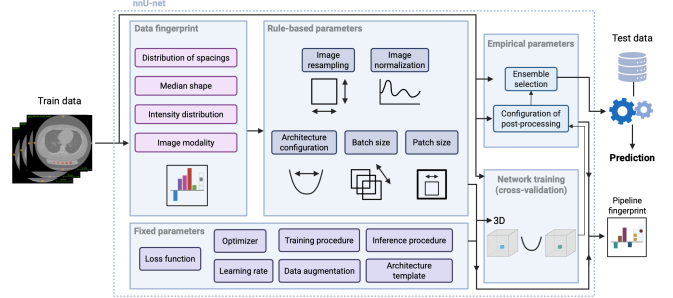


Fig. 2. Overview of the proposed segmentation pipeline. Starting from an input CT scan, the workflow includes dataset planning and preprocessing with nnU-Net v2, model training and inference, and generation of the final binary rod mask. Quantitative evaluation is then performed using overlap and boundary-based metrics, including Dice Similarity Coefficient and Intersection-over-Union. (Figure created by the author, adapted from [2]).

Table 1. Dice similarity coefficient (DSC) and intersection-over-union (IoU) across five-fold cross-validation. Fold-wise results and summary statistics (mean $\pm$ standard deviation) quantify segmentation accuracy and performance variability of the proposed model.

Metric	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Mean $\pm$ SD
DSC	0.907	0.886	0.915	0.907	0.907	0.904 $\pm$ 0.018
IoU	0.831	0.796	0.844	0.831	0.831	0.827 $\pm$ 0.018

delineation of phantom calibration rods despite substantial dataset heterogeneity.

Table 1 reports the fold-wise validation DSC scores and IoU values, along with summary statistics (mean $\pm$ standard deviation), providing a quantitative assessment of both segmentation accuracy and performance variability across folds.

In addition, the segmentations predicted from running inference on the model resulted in a mean DSC of **0.872** and a mean IoU of **0.781**. Further, these segmentations have a sensitivity of **0.876**, along with a specificity of **0.999** and an accuracy of **0.999**, although these values are heavily biased by the imbalance between foreground and background voxels. Lastly, Figure 3 presents representative qualitative results, including both successful segmentations and challenging cases across different scan types and artifact conditions.

The results demonstrate that the proposed DL framework achieves high overlap agreement with manual reference masks while maintaining consistent performance across anatomically and technically diverse CT acquisitions.

4. DISCUSSION

These findings support the use of nnU-Net for the automatic binary segmentation of CT phantom calibration rods. A sig-

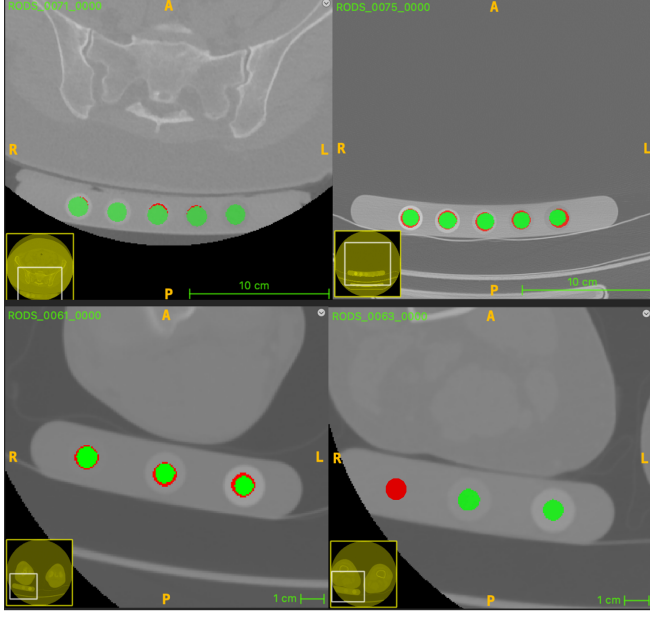


Fig. 3. Qualitative examples of model predictions on heterogeneous CT scans. The figure shows 2D slices of the best segmentation result (top left, DSC=0.942), an average segmentation result (top right, DSC=0.871), a poor segmentation case (bottom left, DSC=0.775), and the worst segmentation case (bottom right, DSC = 0.618) that missed an entire rod, with overlays of predicted masks (green) and true labels or ground truth (red).

nificant advantage of nnU-Net is that it reduces the burden of manual architecture and preprocessing design while still achieving strong segmentation performance across diverse biomedical applications [2]. For this dataset, this property is especially valuable because variability was not limited to only anatomical context but also included scanner vendor, energy, reconstruction kernel, table height, and artifact status. Although a manually tuned pipeline may also achieve useful performance, a self-configuring framework is more reproducible and defensible for the scope of this project.

The present findings are also consistent with broader evidence that U-Net-based segmentation remains highly competitive in medical imaging, particularly when methods are rigorously configured and validated [4, 8]. Rather than pursuing architectural novelty, this work prioritized robustness and methodological suitability.

A likely contributor to the observed performance was the inclusion of heterogeneous scans during training. Prior literature has shown that segmentation performance can be sensitive to acquisition shifts and image quality changes [13]. By incorporating scans with multiple vendors, energies, kernels, anatomical contexts, and error cases, the training data likely exposed the model to a broader range of appearances than a narrowly homogeneous dataset. This may have improved

robustness, although subgroup-specific analyses would be needed to quantitatively verify this effect.

Prior literature also supports the practical relevance of the task. Phantom-based calibration is used to support quantitative CT measurements, and calibration procedures have been shown to reduce systematic measurement error and improve reproducibility in CT systems [14]. In addition, post hoc quantitative CT studies have shown that calibration behaviour can vary with scanner settings and phantom choice [15]. Although this project does not quantify downstream calibration error, these studies indicate that reliable localization of phantom structures is important for subsequent quantitative analysis.

Several limitations should be acknowledged. First, the dataset size was modest, with 60 labelled cases in total. Although five-fold cross-validation partially mitigates this limitation, a relatively small dataset still constrains statistical confidence and may not capture all relevant failure modes. Second, the quantitative analysis was limited to Dice coefficient, which provides a useful overlap-based summary but does not characterize all aspects of segmentation behaviour, such as boundary accuracy or specific error types. Third, the task was defined as binary segmentation of all rods as one class. This is appropriate for the current objective, but it does not address rod identity, individual rod instance separation, or geometry-specific labelling, which may be relevant for some downstream calibration workflows.

The proposed method is a promising alternative to manual segmentation of CT calibration rods. Automated CT segmentation has been shown in other application domains to improve efficiency while preserving accuracy [16]. For phantom rod analysis, reducing manual effort could improve reproducibility and make large-scale or multi-site calibration studies more feasible. Nevertheless, as with other automated segmentation systems, further inspection of outputs remains advisable, especially for scans with strong artifacts, unusual cropping, or uncommon phantom configurations.

5. CONCLUSION

This project developed a DL approach for automated binary segmentation of phantom calibration rods in heterogeneous CT scans. Using nnU-Net v2 and a dataset of 60 labelled scans acquired under variable anatomical and technical conditions, the model achieved validation results of a mean DSC of 0.904 and a mean IoU of 0.827, while test accuracy was slightly lower with a mean DSC of 0.872 and a mean IoU of 0.781.

These findings show that rod segmentation is feasible despite substantial CT heterogeneity. The proposed framework offers a practical and scalable alternative to manual segmentation, with potential value for phantom-based quantitative CT workflows.

6. REFERENCES

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